

ALEX CHEVRIER

Low-Latency C++ Systems Engineer

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SKILLS

Languages: C++23, Java 25, SQL

Low-latency: lock-free SPSC/MPSC, seqlock, mmap, memory ordering,

pthread_setaffinity_np, perf/massif, GBench/GTest, CMake, Conan

Systems: Linux/GCC, Kafka, binary TCP/NIO, Docker, AWS

PROJECTS

Low-Latency Order Book (C++23) | github.com/alchevrier/low-latency-order-book

- N-SPSC MPSC, SOA order book (sorted int64_t tick arrays), seqlock sync, pinned threads (CPU topology analysis, IRQ affinity steering). Median 2.5 ns / p99.9 7.9 ns; HT sibling interference: -67% p99.9 from core selection alone. Zero heap (massif). 10 ADRs.

Low-Latency Log Engine (C++23) | github.com/alchevrier/low-latency-log-engine

- mmap-backed SPSC log; median 4.3 ns / p99 9.5 ns. Zero heap across 1M iterations. C++23 concepts. 13 ADRs.

Mini-Kafka (Java 25) | github.com/alchevrier/distributed-messaging

- Append-only log, binary TCP/NIO, Raft consensus. Off-heap MemorySegment: p99.99 -77% GC (JFR-confirmed).

EXPERIENCE

Morgan Stanley, Hong Kong — Software Engineer (Contract) Nov 2025–Present

- Designed greenfield messaging service; reverse-engineered Python batch system into streaming Kafka pipeline.
- Maintain production Kafka infrastructure across isolated data centres.

We+, Hong Kong — Software Engineer Sept 2023–Nov 2025

- Migrated highest number of microservices to Java 21 across French investment bank FX platform.
- Rebuilt RHEL9 environments; produced runbooks eliminating external dependency.

LandyTech, London — Software Engineer Oct 2019–Aug 2023

- Monolith-to-microservices migration; automated risk data pipeline.

Redbite Solutions, Cambridge — Software Engineer Nov 2016–Oct 2019

- Built web application from scratch; implemented OAuth2 authentication layer.

EDUCATION

ENSEIRB-MATMECA, Bordeaux — M.Sc. Computer Science 2013–2016

Admitted by competitive national examination (Grande École entrance).